SALTDRAG · PROJECT PLAN

SaltDrag project plan

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Sets per pan decided (D25), 2026-10-25
0.1	2026-10-24	Amish Chadha	First project plan (kit 1.8 upgrade); stage P3 reached; references of every document collected

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SaltDrag is at stage P3 (design outputs, TRL 3), the portfolio's stage cap; the next gate is P4 (verification, TRL 4), which is on hold until Amish lifts the cap. Decisions requested now: the concept choice between options A, B and C, which score within 10 % (SDG-OPT-001), and whether one set per family pan is enough (SDG-DEC-001, open decisions 1 and 2). Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Concept and STEM check	Design lead	SDG-STM-001, SDG-OPT-001, SDG-DDR-003	Done at P3; STEM check open on five field-data rows; concept choice awaiting Amish
Requirements and traceability	Design lead	SDG-URS-001, SDG-PRS-001, SDG-RTM-001	Done at P3; PR-005 not verified
Calculations	Design lead	SDG-CAL-001 v0.3	Done at P3
Detailed CAD and drawings	Design lead	`cad/src/model.py` (three options at D3), STEP files, SDG-DWG-001 and SDG-DWG-101 to 113	Done at P3
Build plan and BOM	Design lead	SDG-BLD-001, bom/bom.csv	Done at P3
Regulatory	Co-design partner, with the project lead	Written answer from the forest department (SDG-REG-001 G2)	To start, through the partner
Co-design partner	Project lead	An agreed partner and trial pans	Not started; SEWA is the first candidate to approach
Verification (P4)	Design lead with a fabricator	Build, proof load with CalRig, drag measurement, timed trials	On hold (stage cap P3)
Validation (P5)	Co-design partner	Season trial, dantala comparison, heat and work-rest study	On hold

Resources needed

Table 2. Resources (P4 resources are listed for planning only; nothing is committed while the cap is P3).

Resource	Deliverable	Effort	Estimated cost (USD)
Co-design partner liaison	Partner agreement, trial pans, forest department request	5 to 10 days	to assess
Town fabrication workshop (P4, on hold)	One prototype set	About 4 weeks	2,897 parts and labour (bom/bom.csv)
CalRig proof-load rig access (P4, on hold)	Proof load of rope, capstan and anchors to 4.5 kN	1 day	to assess
Field measurement kit (P4, on hold)	Spring scale to 500 N, short trial rake, soil pull test	2 days on site	to assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB, STM, OPT, REG, FEA, PLN	Reached (documents complete 2026-10-24)
P1 Definition	URS, PRS draft, RSK with hazard analysis, MKT	Reached 2026-10-24
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept selection DDR	Reached 2026-10-24; STEM check open on field-data rows
P3 Design outputs	Model at D3, STEP, drawings, priced BOM, BLD, DEC, design FMEA, traceability	Reached 2026-10-24 (kit 1.9.1)
P4 Verification	Prototype built; proof load; drag, crank force and set-up time measured; RSK updated	On hold (stage cap P3)
P5 Validation	Season trial with a partner; dantala comparison (PR-005); forest department agreement	On hold
P6 Commercialization	Only with a named deployment partner	On hold

Next steps and decision requested

1. Amish: choose among options A, B and C, or ask for more evidence first (SDG-DEC-001, open decision 1; SDG-DDR-003). The repository stays on option A meanwhile.
2. Decided by Amish on 2026-10-25 (SDG-DEC-001 D25): one set per family pan is the design case. With the partner: ask how long the gathering season really is and what share of the harvest a family would gather with the tool, before any scope change.
3. Amish: review the eight fixes made under the standing instruction (SDG-DEC-001 D17 to D24), any of which he can reverse.
4. Approach SEWA and one other producer group with the costed set, the safety case and the co-design checklist (SDG-PRB-001).
5. Through the partner, ask the forest department in writing whether a hand capstan is allowed (SDG-REG-001 G2).

6. When the cap allows P4: measure the drag per metre and the dantala rate on a partner's pan before building, then build and proof-load one set.

References

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